HOMEWORK 2-NORTHWESTERN SUMMER SCHOOL

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Consider single index models. Changing the activation changes the information exponent we saw. Consider the problem of finding the best one. Let $\tau : \mathbb{R} \to \mathbb{R}$. We will consider *label transformations,* that is, consider the single index model with data $(\tau(y), a)$. Let the information exponent of the corresponding problem be $\mathcal{I}(\tau)$.

Definition. Given an activation function σ, the *generative exponent* is the minimal $\mathcal{I}$ over L^2transformations,

$$\mathcal{G}(\sigma) = \min\{\mathcal{I}(\tau) : \tau(y) \in L^2\}$$

Let p be a non-constant polynomial. We will show that $\mathcal{G}(p) \leq 2$.

Exercise 1. Show that if $\phi(t) = \exp(-x^2/2)/\sqrt{2\pi}$, then for a gaussian random variable Z, $\mathbb{E}Z\mathbb{1}_{Z\geq t} = \phi(t)$ and $\mathbb{E}Z\mathbb{1}_{Z\leq t} = -\phi(t)$.

Exercise 2. Suppose that p is odd, show that $\mathcal{G}(p) = 1$. [Hint: w.l.o.g. p is monic. Show that for some large enough t_0 we have that $\{p(x) > t\} = \{G > K(t)\}$ for some $K = K(t)$ for all $t \geq t_0$. Use $\tau(y) = \mathbb{1}_{y>c}$.

Exercise 3. Suppose that p is even. Show that $\mathcal{G} = 2$. [Hint: Its clearly at least 2. Now look at $q(x^2) = p(x)$and go from there.]

Exercise 4. What about $\deg p$ even but p not even? Show that $\mathcal{G} = 1$.

Exercise 5. Give an elementary proof of the **strong** functional law of large numbers for SGD in the classical asymptotics regime. That is, the evolution should converge almost surely to population gradient flow. Prove this under enough moments of the data distribution, probably 2 is enough, but you can assume what you want, optimize for the simplest proof.

Exercise 6. Consider binary logistic regression, that is, $(y, X)$is such that $y \sim Ber(1/2)$ and

$$X \sim \begin{cases} \mathcal{N}(\mu, I_d) & y = 0 \\ \mathcal{N}(-\mu, I_d) & y = 1 \end{cases}$$

for some $\mu \in \mathbb{R}^d$ with $||\mu|| = 1$. Take as loss the binary cross entropy loss

$$L(y, x) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$$

where $\hat{y} = (1 + e^{-x\cdot\theta})^{-1}$ is the sigmoid activation. What are the natural summary statistics? Show that this is asymptotically closable at learning rate $\delta = O(1/d)$ for this choice.

Exercise 7. Extend this to multi-class logistic regression. $y \in \{e_1, ..., e_k\}$ and

$$X \sim \begin{cases} \mathcal{N}(e_a, I_d) & y = e_a \end{cases}$$

with the cross entropy loss corresponding to the soft max : $\theta = (\theta_a)_{a\in[k]}$

$$(\hat{y}(x))_b = \exp(\theta_a \cdot x)/\sum_b \exp(\theta_b \cdot x)$$